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The Currency of Reciprocity: Gift Exchange in the Workplace[†]

By SEBASTIAN KUBE, MICHEL ANDRÉ MARÉCHAL, AND CLEMENS PUPPE*

The psychological impact of providing tangible or intangible gifts to employees is likely to depend not only on the magnitude of the gifts but also on the gifts being seen as (...) costly to the donor in terms of time or effort.

— James N. Baron and David M. Kreps (1999)

How can firms motivate their employees to provide effort above the minimal level? This question is of great importance for both theorists and practitioners. Assuming that workers strictly pursue what is in their material self-interest, a large theoretical literature explores how explicit and implicit contracts can be designed so that the workers' interests are aligned with the firm's objectives (see MacLeod 2007, Prendergast 1999, or Gibbons 1998). A different strand of literature, based on sociological and psychological insights, questions the assumption of pure self-interest, underlining the importance of reciprocity¹ in the presence of incomplete contracts (see Fehr, Goette, and Zehnder 2009). According to this view, firms might achieve higher profits by treating their workforce kindly (e.g., paying fair wages) because workers reciprocate positively to "gifts" and return favors by exerting higher effort (Akerlof 1982). The empirical evidence on gift exchange is mixed. While the results from laboratory experiments are broadly conclusive and suggest that fixed wages influence effort positively (e.g., Fehr, Kirchsteiger, and Riedl 1993; Hannan, Kagel, and Moser 2002; Charness 2004), recent field experiments provide only weak or moderate support for positive reciprocity (e.g., Gneezy and List 2006; Kube, Maréchal, and Puppe forthcoming; Cohn, Fehr, and Goette 2009). Both types

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¹ By reciprocity, we refer to the behavioral phenomenon of people responding toward (un)kind treatment likewise, even in the absence of reputational concerns. Economic theories formalize reciprocal behavior by incorporating the distribution of outcomes, the perceived kindness of intentions, or simply emotional states as arguments into the individual utility function (see Charness and Rabin 2002; Falk and Fischbacher 2006; Rabin 1993; Dufwenberg and Kirchsteiger 2004; or Cox, Friedman, and Gjerstad 2007).

of approaches, however, have focused on monetary gifts and paid little attention to the nature of gifts.²

This paper fills this gap and analyzes how strongly workers reciprocate nonmonetary and monetary gifts with higher productivity. For this purpose, we conducted a controlled field experiment in a naturally occurring work environment. We recruited workers to catalog the books from a library for a limited time, excluding any possibility of reemployment. The job was announced with an hourly wage of 12 euros—the amount actually paid out in our benchmark treatment. In our cash treatment, the workers received a monetary gift in the form of a 20 percent wage increase. In the bottle treatment, we gave workers a thermos bottle of equivalent monetary value.³ The results show that the nature of gifts crucially determines the prevalence and strength of reciprocal behavior. The cash gift had no significant impact on workers' productivity. The bottle, however, resulted, on average, in a 25 percent higher work performance, outweighing the percentage increase in workers' compensation. We replicated the results from our bottle treatment with a control treatment where we explicitly mentioned the bottle's market price. Biased beliefs about the gift's market price thus cannot account for the differences between the cash and the bottle treatments.

We further show that preferences in favor of the nonmonetary gift do not drive our results. Almost all workers preferred the money in an additional treatment where they could choose between receiving cash or the bottle. Strikingly, average work productivity was as high as if workers received the bottle and thus significantly higher than when they only received the money (without having the choice). The latter results are consistent with the common saying that it is the thought—i.e., the time and effort invested into the gift—that counts (see Baron and Kreps 1999; Robben and Verhallen 1994; Webley, Lea, and Portalska 1983). Workers did not choose the bottle, but the employer still incurred the time and effort for choosing, buying, and wrapping the gift. To provide a more direct test whether time and effort matter, we conducted a final treatment. In this treatment the employer gave the workers money in the form of an origami (i.e., the money was artistically folded and wrapped). The origami was identical to the cash gift, except that the employer had invested more time and effort into the gift. The results show that workers reciprocated the origami by producing 30 percent more output relative to the baseline.

The remainder of this paper is organized as follows: We outline the experimental design in Section I and present the empirical results in Section II. Finally, we discuss how our results relate to the theoretical literature on gift exchange and conclude the paper in Section III.

I. Experimental Design

The libraries of two economic chairs at a German university had to be catalogued. We used this opportunity to run a field experiment and recruited workers from all

²See Falk (2007) and Maréchal and Thöni (2010) for field experiments on nonmonetary gift exchange in other contexts.

³The gift came as a surprise for the workers and was not tied to performance. See Jeffrey (2009) and Eriksson and Villeval (2010) for laboratory studies analyzing performance-contingent nonmonetary incentives. See also Kosfeld and Neckermann (2011) for a field experiment studying the effects of symbolic awards.

over the campus using posters. The announcement read that it was a one-time job opportunity for half a day (3 hours), and that pay would amount to 12 euros per hour. We emphasized the one-shot nature of this job offer in order to rule out reputational concerns, which are inherent in ongoing relations. The hourly wage of 12 euros served as a common reference point. A large number of candidates applied during the announcement phase. A research assistant randomly picked subjects from the list of applicants. They were invited with an e-mail and asked to confirm the starting date, reminding them that the job would pay 12 euro per hour.

Upon arriving on the working day, workers were separated from each other and placed in different rooms in front of a computer (with internet browser) and a table with a random selection of books. Their task was to enter each book's author(s), title, publisher, year, and ISBN number into an electronic database. The computer application recorded the time of each log, allowing us to reconstruct the number of characters each person entered over time, without having to monitor work performance explicitly.⁴ A research assistant explained the task, strictly following a protocol.⁵ The workers were allowed to take a break whenever necessary. This data entry task is well suited for our experiment and is frequently used in field experiments because it allows for a precise measurement of output and quality.⁶ Moreover, the task is relatively simple and can be done in isolation, allowing for more control than is usually available in other field settings.

Before workers actually started performing their task, the different treatments were announced: we reminded the workers of their hourly wage and informed them about any additional payments or benefits. The latter additional payments and benefits were the only difference between the treatments. Altogether, we conducted six treatments.

In our benchmark treatment *Baseline*, the workers received 12 euros per hour in cash at the end of the working day, without any additional benefits. In treatment *Money*, total wages were unexpectedly raised by roughly 20 percent by paying an additional fixed amount of 7 euros. In treatment *Bottle*, instead of the pay raise, workers received a thermos bottle worth 7 euros, which was wrapped in a transparent gift paper (see left photo in Figure 1). While the bottle was handed over to the workers immediately with the announcement, the 7 euro cash gift was given together with the regular wage at the end of the employment. In order to account for this potential timing confound, we ran an additional control treatment *MoneyUpfront*, where the 7 euro cash gift was paid out immediately after the announcement. Paying 7 euro at the end together with the hourly wage seems more natural, but *MoneyUpfront* more closely resembles treatment *Bottle* with respect to the timing of the gift. We found no significant performance differences, however, between *Money* and *MoneyUpfront* (Wilcoxon rank sum test, $p = 0.756$, two-sided). We therefore pooled them in the following analysis and refer to them together as treatment *Money*.

⁴See Figure A2 in the Appendix for a screen shot of the computer application.

⁵Within each wave of experiments, all workers interacted with the same female research assistant, preventing potential confounding experimenter effects. The research assistants neither knew the purpose of the study nor the reason for the treatment variations.

⁶See Gneezy and List (2006); Kube, Maréchal, and Puppe (forthcoming); Kosfeld and Neckermann (2011); and Hennig-Schmidt, Rockenbach, and Sadrieh (2010) for some recent examples.



FIGURE 1. GIFTS IN KIND: BOTTLE AND ORIGAMI

Notes: The first photo on the left depicts how the bottle was presented. In treatment *PriceTag* the 7 euro price tag was left visible at the bottom of the bottle. The other two photos contain the 7 euro origami in and outside of the envelope.

In order to control for workers' perceptions about the actual price of the bottle, we ran treatment *PriceTag*. *PriceTag* was analogous to *Bottle*, except that we explicitly mentioned the bottle's market price and marked it with a corresponding price tag. A comparison of the treatments *PriceTag* and *Bottle* allowed us to assess the robustness of our results with regard to the uncertainty of the actual price of the gift.

In treatment *Choice*, workers could choose between receiving 7 euros in cash or the bottle. We presented the bottle in exactly the same way as in *PriceTag*, ensuring that every worker knew that the two options were equivalent in monetary terms. Treatment *Choice* served two purposes. First, it allowed us to elicit revealed preferences for receiving cash or the bottle. This will illustrate whether a preference in favor of one of the different gifts drives treatment differences between *Money* and *Bottle*. Second, it allows us to test the importance of correctly guessing the recipients' preferences (see Prendergast and Stole 2001). By providing the alternative of 7 euros in cash, the employer sends a weaker signal about his knowledge of the recipient's tastes.

Finally, we used treatment *Origami* to test whether the time and effort invested in the provision of gifts matters. Workers received an origami shirt, folded out of a five euro bill, and a two euro coin glued together on a plain postcard. The gift card was also wrapped in a transparent envelope (see the second and third photos from the left in Figure 1). Treatment *Origami* mirrored treatment *Money*, except that the employer invested more time and effort in the cash gift.

All types of gifts (7 euros in form of cash, bottle, or origami) were announced in the same way: "We have a further small gift to thank you: You will also receive [type of gift(s)]." Table A4 in the Appendix provides an overview of the different treatments with a translation of their announcements.

We conducted two waves of experiments, one in May 2007 and the other in July and August 2010. The experiments took place over 12- and 15-day periods, respectively, with up to 6 workers per day. The first wave included treatments *Baseline*, *Money*, *Bottle*, and *PriceTag*. The second wave included *MoneyUpfront*, *Choice*, *Origami*, and an additional *Baseline* treatment. We conducted a second *Baseline* treatment in

order to control for temporal productivity differences between the two waves. We found no significant productivity differences between the two benchmark treatments (rank sum test, $p = 0.373$, two-sided) and therefore pooled the data in the analysis. All parametric regression models contain a dummy variable controlling for the wave.

The treatments were randomized over time slots and weekdays to avoid treatment effects from being confounded by general productivity shocks occurring at different times of the day or weekdays. The allocation of workers to the various treatment groups was randomized as well. We further took great care to avoid any treatment contamination through social interaction and requested workers to arrive sequentially at different times (three workers each in the morning and in the afternoon) and seated them in separate rooms. Moreover, we did not tell them that we had employed other workers. The invited workers were randomly selected from the pool of applicants, which consisted of about 300 applicants in the first and 110 in the second wave. None of the workers from the second wave had participated in the first wave. Roughly 11 percent of the invited workers failed to show up at the scheduled time. We had a total of 35 workers in the Baseline (17 in Baseline I and 18 in Baseline II), 34 in Money (16 in Money and 18 in MoneyUpfront), 15 in Bottle, 15 in PriceTag, 18 in Origami, and 22 in Choice.

After three hours elapsed, all workers completed a short employee questionnaire and received their total wages. In order to observe them in a natural environment, the workers were not told that they were participating in an experiment.

II. Results

The number of characters entered measures workers' productivity precisely and is considered as our main outcome variable for the subsequent analysis.⁷ Figure 2 depicts the development of output over time for treatments Money and Bottle in comparison with treatment Baseline. Consistent with previous field experiments involving monetary gifts, a plain wage increase of roughly 20 percent had only a moderate impact on productivity: compared to the benchmark treatment, the average number of characters entered was approximately five percent higher in treatment Money. As indicated in Table 1, this difference does not reach statistical significance (Wilcoxon rank sum test: $p = 0.670$). Result 1 summarizes this behavioral regularity:

RESULT 1:

The unexpected 20 percent fixed pay raise in treatment Money increased workers' productivity by 5 percent on average. This effect, however, does not reach statistical significance.

The results from treatment Bottle, on the other hand, paint a different picture. Workers entered, on average, roughly 25 percent more characters than in Baseline. Moreover, as illustrated in Figure 2, panel B, this treatment effect remained large over the entire duration of the experiment. The gift implied an increase in

⁷We focus on output quantity first and postpone the analysis of the quality dimension of work performance to the end of this section.

TABLE 1— AVERAGE TREATMENT EFFECTS: NUMBER OF CHARACTERS ENTERED

	Baseline	Money	Bottle	PriceTag	Choice
Money	+5.2 percent				
Bottle	+24.8 percent ***	+18.7 percent **			
PriceTag	+21.4 percent ***	+15.5 percent **	−2.7 percent		
Choice	+24.5 percent ***	+18.4 percent **	−0.2 percent	+2.5 percent	
Origami	+29.3 percent ***	+23.0 percent **	+3.6 percent	+6.5 percent	+3.8 percent

Notes: This table reports average treatment effects (in percentage) for all treatment comparisons (i.e., treatments indicated in the first column are compared with those in the first row). The outcome variable is the number of characters entered as a performance measure. Significance levels from a nonparametric (two-sided) Wilcoxon rank sum test for the null hypothesis of equal output between treatments are denoted as follows:

- *** Significant at the 1 percent level.
- ** Significant at the 5 percent level.
- * Significant at the 10 percent level.

workers’ compensation by only 20 percent. The elasticity of output with respect to compensation amounts to remarkable 1.23. Table 1 shows that the gift-exchange effect is also significant from a statistical point of view. Using Wilcoxon rank sum tests, the hypotheses of identical productivity between treatments Bottle and Baseline (as well as between Bottle and Money) are rejected ($p < 0.01$, respectively $p < 0.05$). The main findings are summarized in our second result:

RESULT 2:

In contrast to the pay raise, a gift in kind of equivalent monetary value resulted in a statistically significant 25 percent productivity gain. This effect was larger than the relative increase in labor compensation.

Understanding the Currency of Reciprocity.—Given that there was no price tag on the bottle, most workers probably were unaware of its exact market value. Workers might have systematically overestimated the market value, which could potentially explain the larger treatment effect in Bottle relative to Money. Treatment PriceTag allows us to test whether the uncertainty with respect to the gift’s market price drives the effect. Given that we communicated the gift’s price, output should have been lower in treatment PriceTag than in Bottle if workers reciprocated only on the basis of monetary considerations and if they overestimated the gift’s price. The performance pattern in Figure 3 reveals, however, that treatment PriceTag closely replicated the results from treatment Bottle. Workers were slightly less productive in PriceTag than in Bottle—measured output was 2.7 percent lower. This difference, however, does not reach statistical significance (Wilcoxon rank sum test: $p = 0.663$).

Similar to Bottle, treatment PriceTag resulted in a 21 percent higher output compared to the benchmark treatment ($p = 0.005$). These productivity gains were still slightly larger than the relative increase in compensation of the workers. We summarize the results as follows:

RESULT 3:

We replicated Result 2 with treatment PriceTag. Workers produced almost an equal output in treatments PriceTag and Bottle. In comparison with Baseline, treatment PriceTag resulted in a 21 percent increase in productivity. The uncertainty

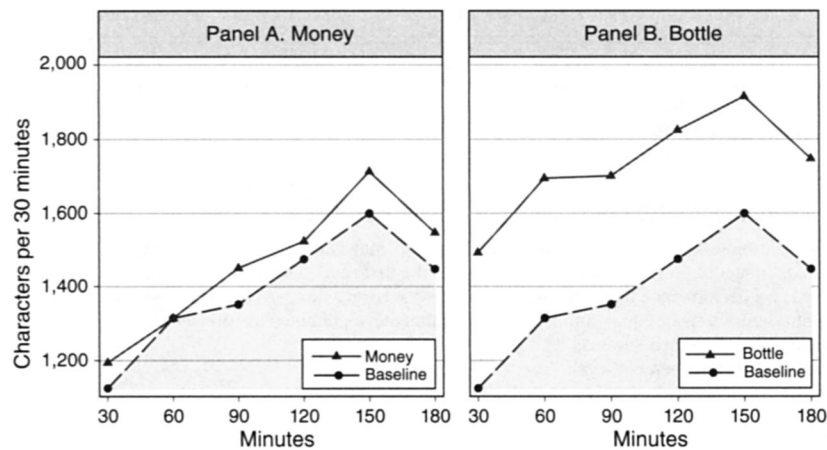


FIGURE 2. MONEY VERSUS BOTTLE

Note: This figure depicts the average number of characters entered per 30 minutes for treatments Money and Bottle, as well as work performance in the benchmark treatment Baseline.

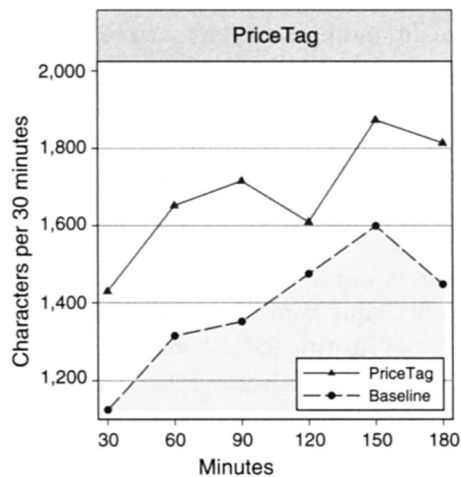


FIGURE 3. PRICETAG

Note: This figure depicts the average number of characters entered per 30 minutes for treatment PriceTag and the Baseline.

concerning the exact market price of the gift in kind thus failed to account for the treatment effects.

A second important question is whether the workers preferred receiving the bottle rather than its cash equivalent. In treatment Choice, we offered workers the choice between receiving an additional 7 euros in cash or in the form of the bottle. The bottle was presented in the same way as in treatment PriceTag—i.e., all workers knew that the bottle was worth 7 euros. Panel A of Figure 4 shows that the vast majority of workers—18 out of the 22 workers—opted for 7 euros in cash.

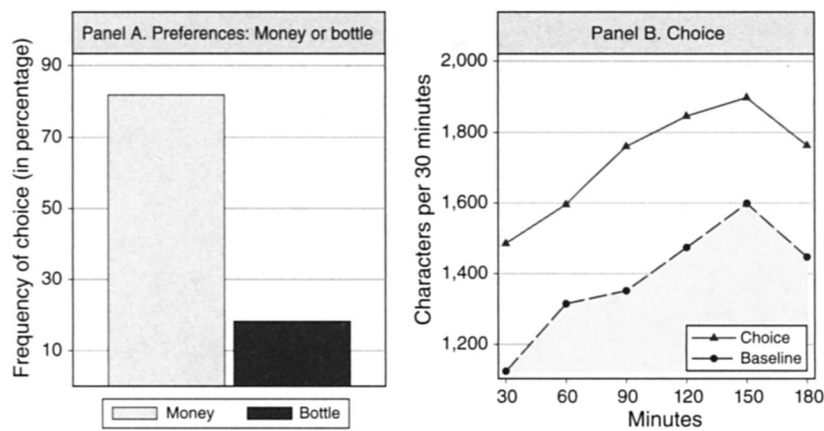


FIGURE 4. CHOICE

Notes: Panel A compares the frequency of choice of the bottle (worth 7 euros) and the 7 euros in cash. Panel B depicts the average number of characters entered per 30 minutes for treatment Choice and work performance in the Baseline.

We are able to reject the hypothesis that workers were drawn from a population in which preferences for cash and the bottle are equiprobable (binomial test, two-sided $p = 0.004$).⁸ We thus conclude:

RESULT 4:

When workers were given the choice between receiving a cash gift of 7 euros or a bottle of equivalent value, more than 80 percent chose the cash gift. The gift in kind thus is unlikely to correspond to its recipients’ preferences.

Panel B of Figure 4 shows how work performance developed over time in treatments Choice and Baseline.⁹ Output was about 25 percent larger in treatment Choice than in Baseline (Wilcoxon rank sum test: $p = 0.006$, see Table 1). This treatment effect is of almost identical magnitude as in treatment Bottle. Moreover, performance was around 18 percent higher than in treatment Money ($p = 0.038$; see Table 1). This result seems surprising, given that almost all workers chose the same gift as in treatment Money. We summarize our results:

RESULT 5:

Despite the fact that almost all workers opted for 7 euros in cash, workers’ output was substantially higher in treatment Choice than in Baseline and Money.

⁸ An earlier version of this paper included additional results from an experiment where subjects in an unrelated lab experiment could actually choose between receiving 7 euros or the bottle in addition to their other earnings. The results were very similar: 159 out of 172 subjects (92.4 percent) opted for 7 euros in cash rather than the thermos of equivalent value (binomial test, two-sided $p < 0.0001$).

⁹ We pooled the 4 workers who took the bottle with the other 18 workers. The treatment effect thus measures the effect of receiving the choice between an additional 7 euros and the bottle. If we were to condition on the workers’ actual choice, we would face potential selection effects. The results are robust, however, if we exclude the four workers who chose the bottle from the analysis.

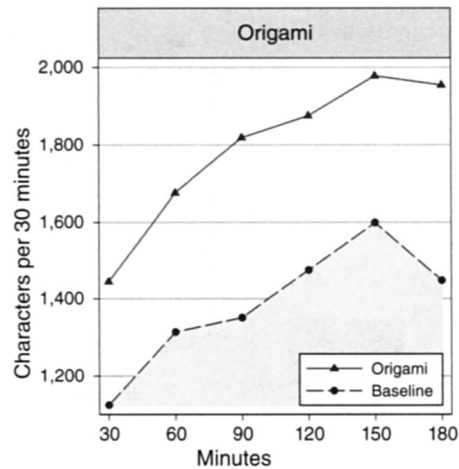


FIGURE 5. ORIGAMI

Note: This figure depicts the average number of characters entered per 30 minutes for treatment Origami and work performance in the Baseline.

Simply offering the bottle was enough to trigger reciprocal reactions—even if the workers did not choose the bottle. Together, Results 5 and 1 are consistent with the common saying “It’s the thought that counts.” One possible explanation for these results is that the time and effort the donor invests in a gift—and not the gift per se—matters. If this explanation indeed drives the results, we should be able to trigger reciprocal responses with money too, provided we invest more time and effort into the cash gift. This is what we did in treatment Origami.

As depicted in Figure 5, treatment Origami had a similar treatment effect as the bottle. Output was almost 30 percent higher in Origami than in Baseline, which is significant from a statistical point of view ($p = 0.001$; see Table 1). This treatment effect was even slightly larger than the effect of the bottle, but the difference does not reach statistical significance ($p = 0.539$). In contrast, productivity was 23 percent higher than in treatment Money ($p = 0.012$). We summarize these observations in the following result:

RESULT 6:

The origami—i.e., an artistically folded and wrapped cash gift—resulted in a statistically and economically significant productivity gain.

Robustness Checks.—The cumulative distribution functions in Figure A1 in the Appendix show that one or two single workers did not drive our results; instead the treatment effects reflect broad behavioral phenomena. In comparison with treatment Baseline, the performance distributions in treatments Bottle, PriceTag, Choice, and Origami were clearly shifted toward higher output levels. The cumulative distribution function from treatment Money, however, was closely intertwined with that from Baseline. For example, the share of workers entering 10,000 characters or less was around 40 percent in treatment Origami. In contrast, this share amounted to 80

percent in treatment Baseline. Pairwise Kolmogorov-Smirnov tests suggest that the distributions from all gift treatments were significantly different from the Baseline ($p < 0.05$), except for treatment Money ($p = 0.741$). Moreover, the distribution functions for Bottle, PriceTag, Choice, and Origami were all significantly different from Money ($p < 0.05$).

The previous analysis focused on nonparametric unconditional treatment comparisons. We complemented these results with a regression analysis that allowed us to control for various potential performance influences.¹⁰ For this purpose, we constructed a panel dataset by slicing the data into six 30-minute intervals. The benchmark model had the following specification:

(1)
$$Y_{it} = \alpha + \beta_1 G_i + \beta_2 T_{it} + \gamma X_i + \rho_i + \theta_i + \delta_i + \omega_i + \epsilon_{it},$$

where Y_{it} represents the number of characters entered by worker i in time interval t ; G_i is a vector consisting of dummy variables indicating each of the different gift treatments. Treatment Baseline was omitted from the model and served as the reference category; T_{it} takes values from zero to five, indicating the six time intervals; X_i is a vector containing controls for the workers' age and gender. The wave fixed effect (ρ_i) controls for general performance differences between the two waves of experiments. Furthermore, we included weekday (δ_i) and room (ω_i) fixed effects as well as a dummy variable for sessions conducted in the afternoon (θ_i). We estimated our model using ordinary least squares (OLS). Standard errors were corrected for clustering, accounting for individual dependency of the error term ϵ_{it} over time.

The results from the benchmark model are displayed in column 1 of Table 2. Consistent with the nonparametric analysis, the coefficient for Money is small and statistically insignificant. In contrast, all other gift treatments have large positive and significant coefficient estimates. Furthermore, the Wald tests reported at the bottom of Table 2 suggest that the coefficients for Bottle, PriceTag, Choice, and Origami are significantly different from Money. The results also suggest the presence of a significant learning effect as indicated by the positive time trend. We explored how the treatment effects evolved over time by interacting all treatment dummies with the time trend in the extended model in column 2 of Table 2. The results remain robust and none of the interaction effects is significant, suggesting that treatment effects remained stable over time (see also Figures 2 through 5).

In contrast to the quantity of output, *quality* is more difficult to observe for the employer. An important question is therefore whether the observed productivity gains primarily stemmed from workers producing more low-quality output. In order to test for quality differences, we measured output quality by the ratio of correctly entered books to the total number of books entered.¹¹ With a quality ratio of 81.4 percent, quality was lowest in the benchmark treatment. Apart from the higher quantity of output, workers also provided better-quality output in all gift treatments,

¹⁰ See Table A1 for summary statistics of our control variables.
¹¹ See Hennig-Schmidt, Rockenbach, and Sadrieh (2010) and Kube, Maréchal, and Puppe (forthcoming) for a similar approach. Two research assistants searched for spelling mistakes in the titles (using an automatic spell check program) and ISBN numbers of the books.

TABLE 2—REGRESSION RESULTS

	Entries		Correct entries	
	(1)	(2)	(3)	(4)
Money	40.350 (85.335)	9.284 (84.984)	123.419 (83.760)	74.576 (80.880)
Bottle	369.563*** (108.830)	407.991*** (107.783)	363.303*** (117.977)	416.061*** (117.061)
PriceTag	345.454*** (111.003)	354.163*** (118.220)	411.180*** (108.651)	421.438*** (109.536)
Choice	263.261** (121.061)	279.188** (112.802)	335.597*** (112.297)	306.344*** (107.023)
Origami	316.467*** (117.926)	250.958** (111.468)	436.924*** (111.883)	347.748*** (108.959)
Time	77.654*** (4.824)	74.264*** (8.274)	63.877*** (4.847)	55.347*** (8.331)
2nd wave	73.341 (127.354)	73.341 (127.746)	−15.400 (117.552)	−15.400 (117.914)
Time*Money		12.426 (12.477)		19.537 (12.666)
Time*Bottle		−15.371 (12.026)		−21.103 (13.537)
Time*PriceTag		−3.484 (18.826)		−4.103 (14.812)
Time*Choice		−6.371 (16.668)		11.701 (14.981)
Time*Origami		26.204 (16.267)		35.671* (19.053)
Constant	1,812.104*** (300.568)	1,820.578*** (300.916)	1,481.451*** (298.939)	1,502.776*** (301.266)
Wald tests:				
Bottle = Money	0.003		0.038	
PriceTag = Money	0.014		0.010	
Choice = Money	0.076		0.083	
Origami = Money	0.031		0.012	
Age and gender?	Yes	Yes	Yes	Yes
Weekday FE?	Yes	Yes	Yes	Yes
Afternoon FE?	Yes	Yes	Yes	Yes
Room FE?	Yes	Yes	Yes	Yes
Observations	834			
Number of workers	139			

Notes: This table reports OLS coefficient estimates (standard errors adjusted for clustering are reported in parentheses). Significance levels are denoted as follows:
*** Significant at the 1 percent level.
** Significant at the 5 percent level.
* Significant at the 10 percent level.

including treatment Money. In comparison with the Baseline, the increase in quality was highest in treatment Origami (90.6 percent quality ratio) and lowest in treatment Bottle (83.4 percent quality ratio). Except for treatment Bottle, all quality differences with respect to the Baseline are statistically significant ($p < 0.05$, see Table A5 in the Appendix).

Furthermore, we used the number of characters from correct entries as the dependent measure in columns 3 and 4 of Table 2. This is a composite measure

of work performance, taking both the quantity and the quality dimension of effort into account. All results remain qualitatively unchanged if we use this alternative performance measure. Interestingly, the Time and Origami interaction term is positive and significant on a ten percent level, suggesting that the effect of the Origami even tended to increase over time.

III. Discussion and Conclusion

The results from our field experiment highlight a sharp contrast between nonmonetary and purely monetary gifts. In this section we discuss how our results relate to existing theories of gift exchange and nonmonetary gift giving and conclude the paper with potential avenues for future research. The literature has generally explained gift exchange with outcome- and intention-based theories of social preferences (see Cooper and Kagel, forthcoming, for a survey). These models, however, do not distinguish explicitly between monetary and nonmonetary gifts. An outcome-based model of inequality aversion could explain our results if we assume that workers take the effort and time the employer has invested in the gift into account. These costs would increase outcome inequalities between workers and the employer by reducing the employer's rent.¹² A rigorous formulation of this idea would necessitate the transformation of all goods exchanged in monetary equivalents. Not only must the recipient be able to quantify the effort and time of providing the gifts, he must also form beliefs about the surplus he creates by exerting effort. Moreover, the effort in our case was directed toward the recipient with positive intentions. It seems plausible that meaningless effort would result in no or weaker reciprocal reactions. Models of inequality aversion, however, do not distinguish between the meaningfulness of effort.¹³

The widespread phenomenon of nonmonetary gift giving has frequently puzzled economists because gifts in kind seem less efficient than money. Waldfogel (1993, p. 1328), for example, suggests that holiday gift-giving "destroys between 10 percent and a third of the value of gifts." Several theories explicitly address nonmonetary gift giving. A first class of models proposes that gifts in kind can be of higher subjective value to the recipient than an equivalent cash gift. Search costs as assumed in Kaplan and Ruffle (2009), for example, could imply that the bottle is a product that the workers always wanted to have but have not yet had the opportunity to buy.¹⁴ The results from treatment choice are inconsistent with such an explanation. Almost all workers preferred the money rather than the bottle.

Prendergast and Stole (2001) develop a model showing that gifts in kind allow donors to signal how well they know the recipient's taste or how intimate their relationship is. "An individual who can show that he understands the preferences of his partner is likely to be a more desirable partner than one who has no idea what his partner wants or believes in" (Prendergast and Stole 2001, p. 1795). Two aspects of our results suggest that signaling intimacy does not explain the observed differences between monetary and nonmonetary gifts. First, our choice experiment suggests

¹²This argument implicitly assumes that workers are narrow bracketing; i.e., they ignore the firm's additional sources of income and focus on the rents from bilateral exchange (see Card, DellaVigna, and Malmendier 2011).

¹³For example, inequality aversion would also predict that the employer could induce higher effort by burning his money in front of the employee, or by spending time and effort to buy and wrap an obviously unfriendly gift.

¹⁴See Thaler (2000) for an alternative argument based on self-control issues.

that the bottle did not correspond to the workers' tastes. Second, by offering the choice between money and a bottle, the employer plausibly signals less knowledge about the recipients' tastes. Nevertheless, we observe that workers reciprocated to an equal extent as in treatment Bottle, where they had no choice.

Several scholars have argued that people can use gifts to signal their willingness to cooperate in future relationships (see Camerer 1988; Carmichael and MacLeod 1997; Bolle 2001; and Sozou and Seymour 2005). According to these models, gifts should be costly to the donor and have little value for the recipient. The lower use value ensures that people will not enter into exchange relationships simply to collect valuable gifts. We find that gifts are reciprocated independent of whether the gift is of lower (as in treatment Bottle) or higher use value (as in Choice or Origami where the workers receive money) for the recipient. Furthermore, these models seem to be less relevant for our context, where gifts were offered only after the relationship was established and there was no possibility for future employment.

Lea, Tarpy, and Webley (1987) suggest that one reason for the unacceptability of money as a gift is that it puts an exact monetary value on a relationship. Money could thus potentially reframe a social exchange relationship into a market or commercial relationship. The lab experiments conducted by Heyman and Ariely (2004) provide evidence that is supportive of this argument. In contrast, our results from treatments PriceTag, Choice, and Origami do not corroborate this explanation. The observed treatment effect is equal to that in treatment Bottle, despite salient information about the gift's price.

Finally, a recent model from Ellingsen and Johannesson (2011) proposes that nonmonetary gifts can signal the donors' degree of altruism. The underlying assumption is that in contrast to self-interested donors, altruists find it less burdensome to spend time and effort for other persons. Altruistic donors therefore have a comparative advantage in providing nonmonetary gifts. Assuming that workers are willing to provide more effort for an altruistic employer (e.g., see Ellingsen and Johannesson 2008) our results might be explained by such a model. An explicit test of Ellingsen and Johannesson (2011), however, would require heterogeneity among employers or direct evidence concerning the workers' beliefs about the employer's type.¹⁵

Summing up, our results underline the importance of nonmonetary aspects in employment relations (see also Rhoades and Eisenberger 2002 or Cropanzano and Mitchell 2005) and suggest that reciprocity has its own currency, which probably cannot be measured in terms of monetary value alone. Gift exchange is a more complex phenomenon than previously assumed in the literature. While our study provides suggestive evidence that the investment of time and effort is crucial for successful gift exchange, more theoretical and empirical research is needed to fully understand the exact mechanism of reciprocity. Several aspects and questions are worth further investigations. Employees might reciprocate gifts in other dimensions than productivity, such as absenteeism, retention, or loyalty in general. Do workers still work more if gifts came from someone who did not benefit directly from the extra effort? Moreover, employment is often characterized by long-term relations, which could

¹⁵ An earlier version of this paper included survey evidence suggesting that the bottle is more likely to be considered as a kind action than a pure cash gift. More detailed results are available from the authors upon request.

result in different dynamic effects than those we find in a one-shot relationship. In a dynamic context, workers might become used to receiving gifts on a regular basis and respond less (see also Gneezy and List 2006). Habituation, however, might be inhibited if the timing of giving gifts appears more random and therefore unpredictable. These issues promise to be interesting topics for future research.

APPENDIX

TABLE A1—SUMMARY STATISTICS

Variable	Mean	SD
Age	23.187	2.802
Male	0.446	0.499
Room 1	0.295	0.458
Room 2	0.180	0.385
Room 3	0.165	0.373
Room 4	0.158	0.366
Room 5	0.201	0.403
Afternoon	0.511	0.502
Monday	0.201	0.403
Tuesday	0.194	0.397
Wednesday	0.230	0.422
Thursday	0.187	0.391
Friday	0.187	0.391
Observations	139	

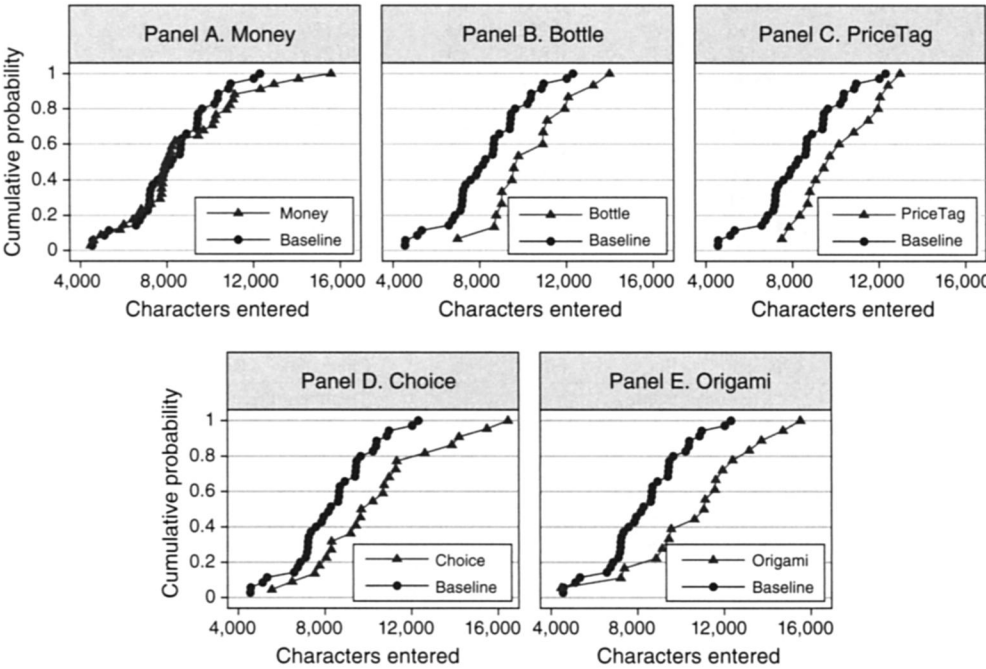


FIGURE A1. CUMULATIVE DISTRIBUTION FUNCTIONS

Note: This figure depicts the cumulative distribution functions of the total number of characters entered for treatments Money, Bottle, PriceTag, Choice, and Origami in comparison with the Baseline treatment.

TABLE A2—DATA OVERVIEW: NUMBER OF CHARACTERS (BOOKS) ENTERED AND QUALITY

ID no.	Total time		Quality ratio	ID no.	Total time		Quality ratio
	Characters	Books			Characters	Books	
Treatment Baseline I				Treatment Money			
1	4,570	44	0.500	36	4,470	50	0.840
2	5,122	55	0.564	37	6,010	71	0.915
3	5,327	42	0.881	38	6,426	60	0.850
4	6,862	75	0.547	39	7,763	77	0.935
5	7,177	76	0.934	40	7,801	77	0.857
6	7,208	78	0.885	41	7,804	80	0.913
7	7,217	75	0.880	42	7,823	82	0.671
8	7,581	66	0.894	43	7,883	87	0.908
9	8,157	57	0.842	44	7,959	84	0.845
10	8,607	93	0.753	45	8,084	76	0.882
11	8,646	105	0.857	46	8,180	91	0.813
12	8,688	97	0.907	47	9,464	100	0.950
13	8,919	95	0.800	48	9,707	96	0.927
14	9,443	99	0.960	49	10,774	94	0.723
15	9,651	106	0.887	50	11,150	112	0.866
16	10,224	112	0.946	51	14,098	148	0.811
17	12,320	136	0.699				
Average	7,983.5	83.0	0.808	Average	8,462.3	86.6	0.857
Treatment Baseline II				Treatment MoneyUpfront			
18	4,552	63	0.825	52	4,611	51	0.765
19	6,575	69	0.652	53	4,941	41	0.854
20	6,741	68	0.926	54	5,840	66	0.833
21	7,114	63	0.794	55	6,686	53	0.906
22	7,247	67	0.791	56	6,816	76	0.934
23	7,348	88	0.841	57	7,266	74	0.811
24	7,847	69	0.899	58	7,690	85	0.835
25	7,936	69	0.899	59	7,761	86	0.849
26	8,277	74	0.770	60	8,254	76	0.961
27	8,658	95	0.853	61	8,397	73	0.959
28	9,396	84	0.952	62	10,126	88	0.932
29	9,422	81	0.815	63	10,218	117	0.940
30	9,464	86	0.802	64	10,289	91	0.989
31	10,365	108	0.611	65	10,928	115	0.913
32	10,392	90	0.889	66	11,082	102	0.922
33	10,865	120	0.825	67	12,345	112	0.938
34	10,964	109	0.817	68	12,965	111	0.559
35	12,034	101	0.812	69	15,603	134	0.933
Average	8,622.1	83.6	0.821	Average	8,989.9	86.2	0.880

TABLE A3—DATA OVERVIEW

ID no.	Total time		Quality ratio	ID no.	Total time		Quality ratio
	Characters	Books			Characters	Books	
Treatment Bottle				Treatment PriceTag			
70	6,979	61	0.721	107	7,503	77	0.909
71	8,671	82	0.683	108	7,836	82	0.939
72	8,756	74	0.919	109	8,332	86	0.884
73	9,018	92	0.783	110	8,701	93	0.957
74	9,027	90	0.744	111	8,804	103	0.854
75	9,492	93	0.903	112	9,066	79	0.873
76	9,581	98	0.857	113	9,449	99	0.889
77	9,796	106	0.868	114	9,729	91	0.703
78	10,922	108	0.778	115	10,164	104	0.663
79	10,939	112	0.866	116	10,846	92	0.957
80	11,123	119	0.731	117	11,517	116	0.836
81	11,936	126	0.865	118	11,972	109	0.844
82	12,102	103	0.951	119	12,059	137	0.949
83	13,254	120	0.933	120	12,436	115	0.896
84	14,011	102	0.902	121	12,994	136	0.904
Average	10,373.8	99.1	0.834	Average	10,093.9	101.3	0.871

(Continued)

TABLE A3—DATA OVERVIEW (Continued)

ID	Total time		Quality	ID	Total time		Quality
no.	Characters	Books	ratio	no.	Characters	Books	ratio
Treatment Choice				Treatment Origami			
85	5,546	49	0.857	122	4,466	56	0.964
86	6,481	58	0.862	123	7,219	74	0.946
87	7,525	78	0.962	124	7,385	81	0.778
88	7,747	81	0.802	125	8,854	98	0.918
89	8,063	72	0.806	126	9,131	74	0.892
90	8,293	96	0.958	127	9,439	94	0.862
91	8,305	91	0.835	128	9,550	85	0.871
92	9,186	84	0.833	129	10,623	96	0.917
93	9,426	76	0.934	130	11,062	96	0.906
94	9,640	104	0.837	131	11,119	99	0.889
95	9,677	93	0.925	132	11,568	103	0.961
96	10,215	93	0.699	133	11,610	118	0.890
97	10,682	87	0.862	134	11,928	101	0.891
98	10,735	108	0.870	135	12,389	114	0.904
99	10,970	102	0.912	136	13,158	149	0.953
100	11,279	124	0.935	137	13,725	153	0.895
101	11,317	120	0.917	138	14,717	130	0.962
102	12,620	139	0.964	139	15,520	147	0.905
103	13,869	115	0.870				
104	14,197	129	0.907				
105	15,482	137	0.825				
106	16,459	143	0.944				
Average	10,350.6	99.0	0.878	Average	10,747.9	103.8	0.906

TABLE A4—TREATMENTS

Treatment	Gift	Announcement	Wave
Money ^a	€7 in cash	“We have a further small gift to thank you: You will also receive €7.”	I&II
Bottle	Thermos bottle (worth: €7)	“We have a further small gift to thank you: You will also receive this thermos bottle.”	I
PriceTag	Thermos bottle with price tag	“We have a further small gift to thank you: You will also receive this thermos bottle worth €7.”	I
Choice	€7 in cash or thermos bottle with price tag	“We have a further small gift to thank you: You will also receive €7. You can choose whether you want to receive the €7 in cash or in form of this thermos bottle worth €7.”	II
Origami	€7 in form of an origami	“We have a further small gift to thank you: You will also receive €7.”	II
Baseline	None	None	I&II

^aTreatment Money (i.e., 7 euros was distributed at the end of the employment) and MoneyUpfront (i.e., 7 euros was distributed at the beginning together with the announcement) are pooled, because there were no significant performance differences between the two treatments.

TABLE A5—AVERAGE TREATMENT EFFECTS: QUALITY RATIO

	Baseline	Money	Bottle	PriceTag	Choice
Money	+6.7 percent **				
Bottle	+2.4 percent	−4.0 percent			
PriceTag	+6.9 percent **	+0.2 percent	+4.4 percent		
Choice	−7.8 percent **	+1.1 percent	+5.3 percent	+0.9 percent	
Origami	+11.2 percent ***	+4.3 percent	+8.6 percent **	+4.0 percent	+3.2 percent

Notes: This table reports average treatment effects (in percentage) for all treatment comparisons (i.e., treatments indicated in the first column are compared with those in the first row). The outcome variable is the quality ratio. Significance levels from a nonparametric (two-sided) Wilcoxon rank sum test for the null hypothesis of equal quality between treatments are denoted as follows:

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

* Significant at the 10 percent level.

FIGURE A2. SCREENSHOT: COMPUTER APPLICATION

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